DP BIOLOGY | SKILLS 3 OF 3

Calculations

How do we turn measurements into evidence?
When is a difference big enough to matter?

All worked examples use invented data. A calculator is allowed in Paper 2.



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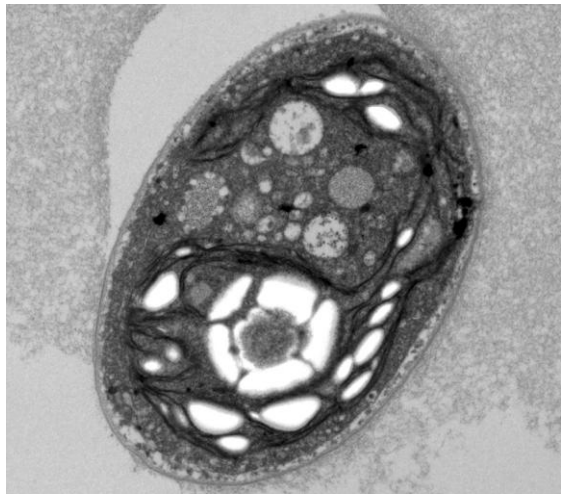
What you will learn



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1 Units and change

Convert units, use standard form, and calculate percentage change



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2 Size and rate

Magnification, actual size, and rates from graphs



Florida Memory, Public domain

3 Describing data

Mean, median, range, standard deviation and error bars



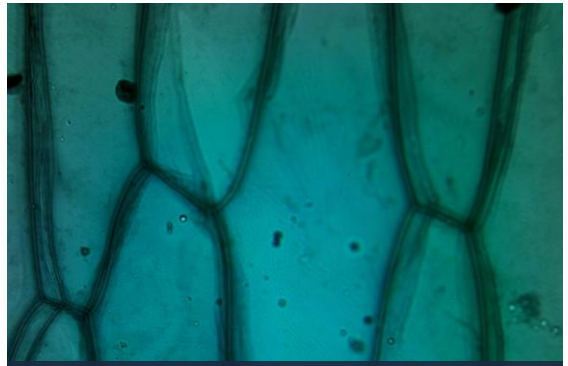
J.Conklin, CC BY 2.5

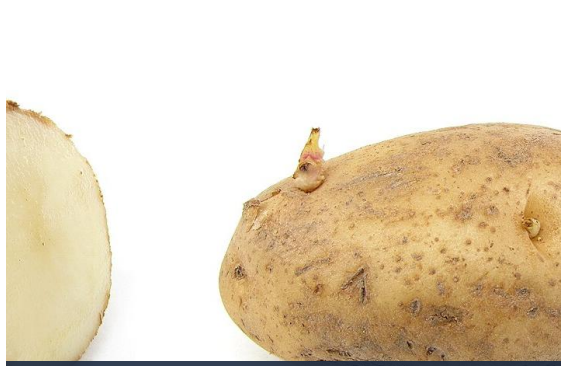
4 Testing data

Correlation, statistical tests and log scales

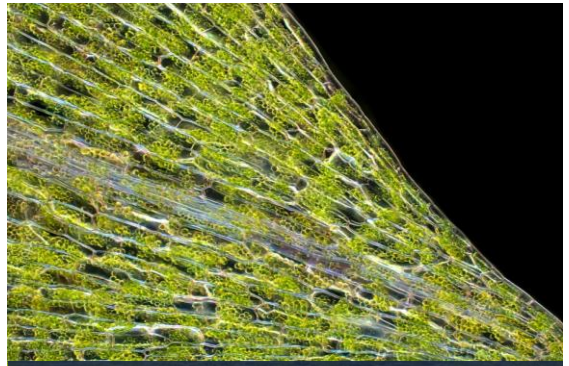
Contents

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Units


Percentage change


Magnification


Rates


Averages


Error bars


Correlation


Tests and logs



01

Units and change

Get the units right before anything else.

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Units and standard form

m — a person

x 1000 →

mm — an ant

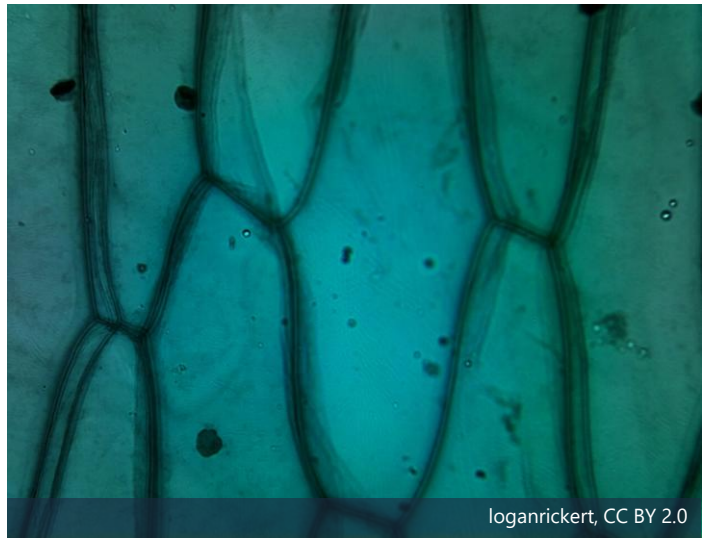
x 1000 →

µm — a cell

x 1000 →

nm — a virus

Moving right, multiply by 1000. Moving left, divide by 1000.



loganrickert, CC BY 2.0

3.2×10^{-6} m

Standard form for 0.0000032 m, which is 3.2 µm.

12.8 s

Mean of 12.4, 13.1 and 12.9 s: three significant figures, like the data.

Rule of thumb

Give a calculated answer to the same number of significant figures as the least precise value you used. The onion cells on the left are about 100 µm, or 0.1 mm, long.

TRY IT

Convert

1. A mitochondrion is 0.0015 mm long. Give its length in µm.
2. A virus is 120 nm across. Give its size in µm.
3. Write 0.00045 m in standard form.

> Hint: Use the ladder: m, mm, µm, nm, each step × 1000.

The answer is on the next slide.

loganrickert, CC BY 2.0

ANSWER

Convert

1. 0.0015 mm × 1000 = 1.5 µm
2. 120 nm / 1000 = 0.12 µm
3. 4.5×10^{-4} m

Compare your working with each step.

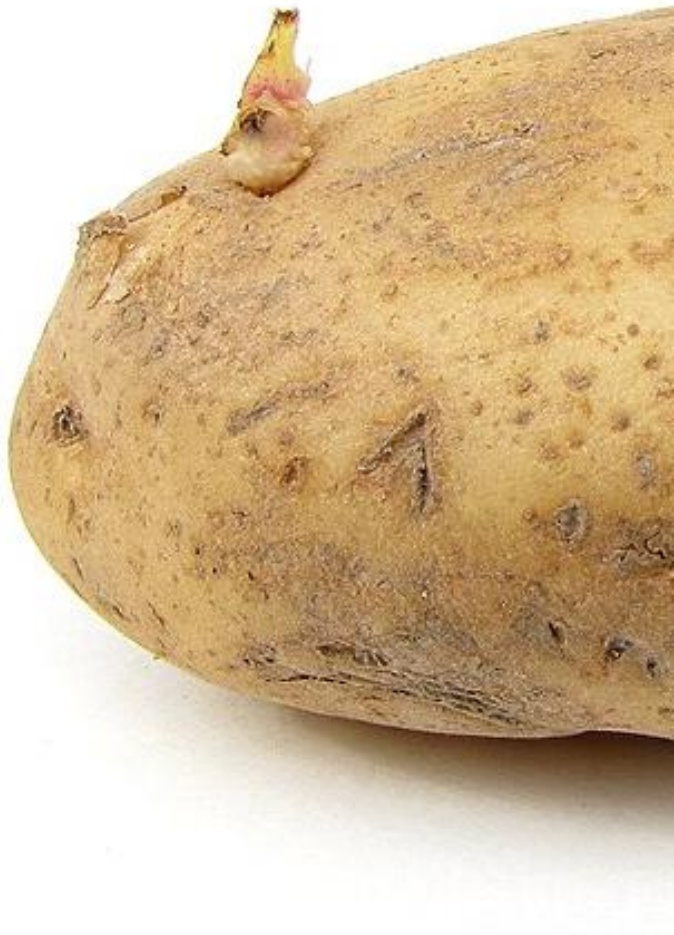
Key point

Check sizes make sense: mitochondria are a few micrometres long, and viruses are far smaller than cells.

Percentage change

Change = (final − initial) / initial × 100

1. Start 2.50 g, end 2.20 g
2. Change = -0.30 g
3. -0.30 / 2.50 × 100 = -12%



Why use it

Potato pieces start at different masses, so percentage change compares them fairly. Keep the sign: a negative value is a loss. Used in D2.3 Water potential.

ZooFari, Public domain

TRY IT

Percentage change

1. A seedling grows from 4.0 cm to 5.8 cm. Calculate the percentage change.
2. A population falls from 250 to 190. Calculate the percentage change.

> Hint: Always divide by the starting value.

The answer is on the next slide.

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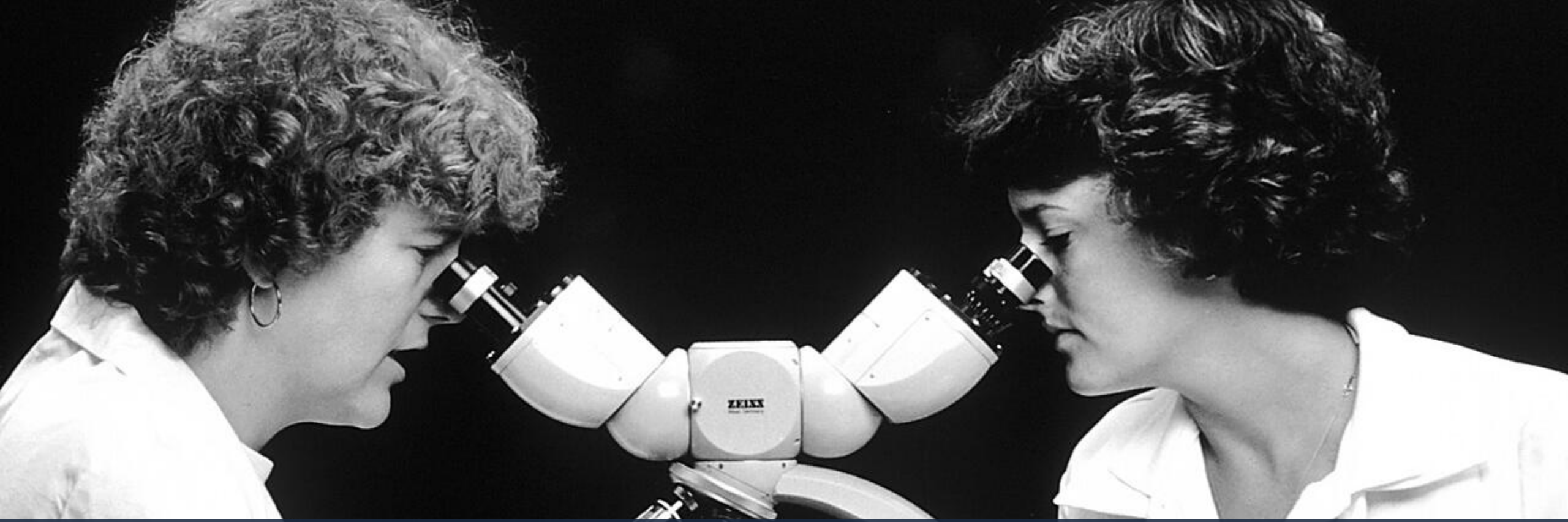
ANSWER

Percentage change

1. (5.8 − 4.0) / 4.0 × 100 = +45%
2. (190 − 250) / 250 × 100 = -24%

Compare your working with each step.

Key point
Dividing by the final value is the most common error.



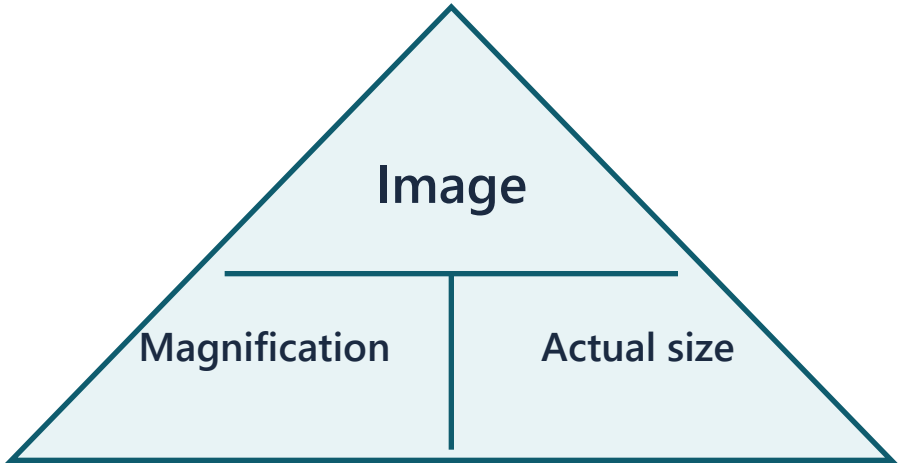
02

Size and rate

How big is it really, and how fast is it changing?

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Magnification and actual size



Cover the one you want: I = M × A, M = I / A, A = I / M

36 000 / 1500 = 24 µm

A cell 36 mm wide in a drawing at x1500. Convert 36 mm to 36 000 µm first.

Scale bars

A 20 mm bar labelled 5 µm: 20 000 / 5 = x4000. Convert to the same unit, then divide.



Mohammad Moniruzzaman, Maria P. Erazo-, CC BY 4.0

TRY IT

Magnification

1. A chloroplast image is 30 mm long at x6000. Calculate its actual length in µm.
2. A cell 80 µm wide appears 40 mm wide in a photo. Calculate the magnification.

> Hint: Convert mm to µm first: multiply by 1000.

The answer is on the next slide.

loganrickert, CC BY 2.0

ANSWER

Magnification

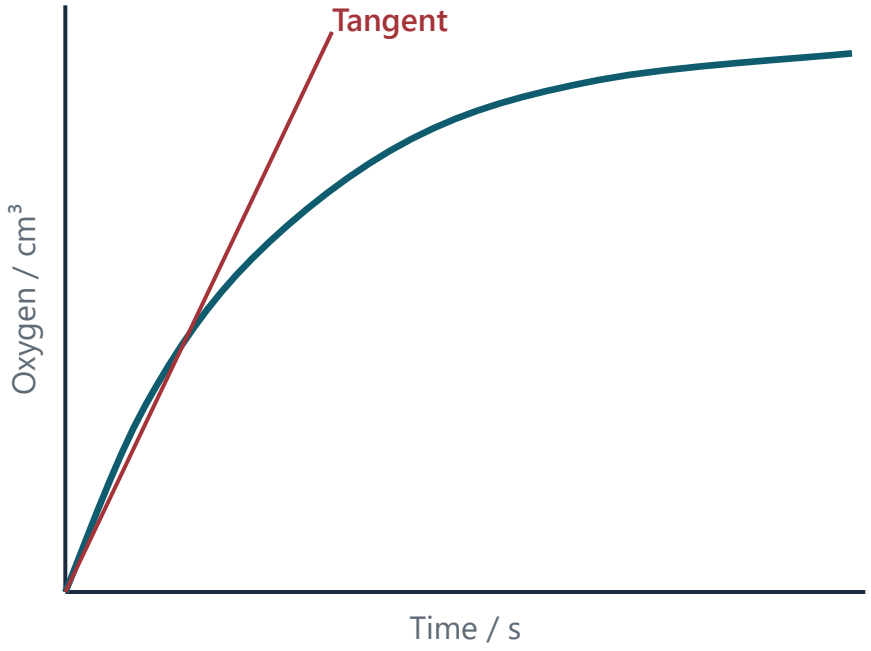
1. 30 mm = 30 000 µm; 30 000 / 6000 = 5 µm
2. 40 mm = 40 000 µm; 40 000 / 80 = x500

Compare your working with each step.

Key point

Magnification has no units, but it is written with a × sign.

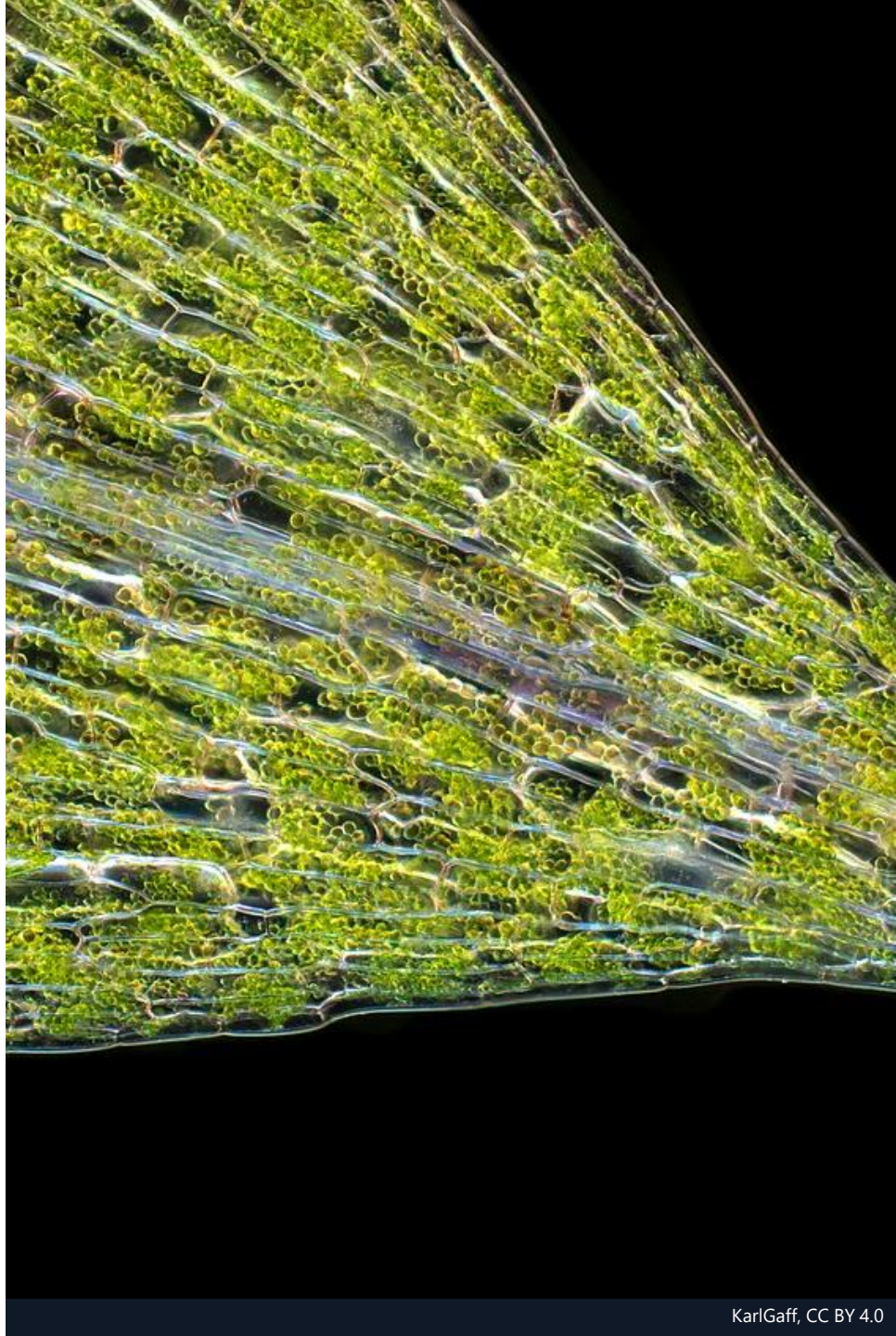
Rates and gradients



- Rate = change / time
- Rate is the **gradient**
- Curved line: draw a **tangent**

$18 / 60 = 0.30\ cm^3\ s^{-1}$

Mean rate when 18 cm^3 of oxygen is released in 60 s. For a reaction timed to an end point, rate = 1 / time.



KarlGaff, CC BY 4.0



03

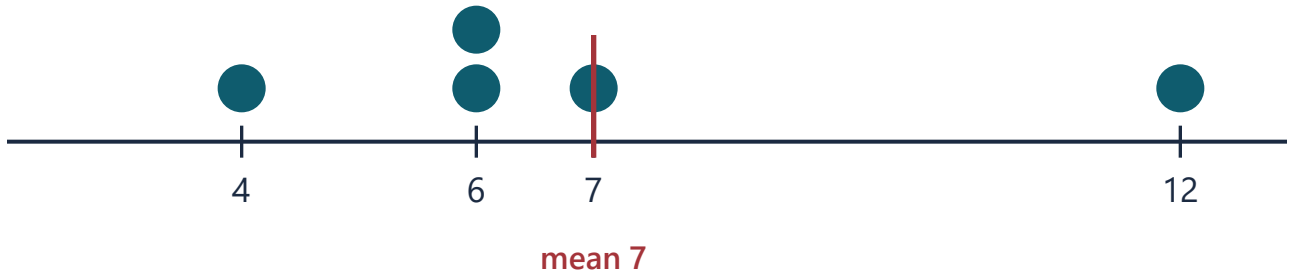
Describing data

One number for the middle, one for the spread.

Florida Memory, Public domain

Mean, median, mode and range

Reaction times / s: 4, 6, 6, 7, 12



Mean

35 / 5 = 7 s

Median

Middle value: 6 s

Mode

Most common: 6 s

Range

12 − 4 = 8 s

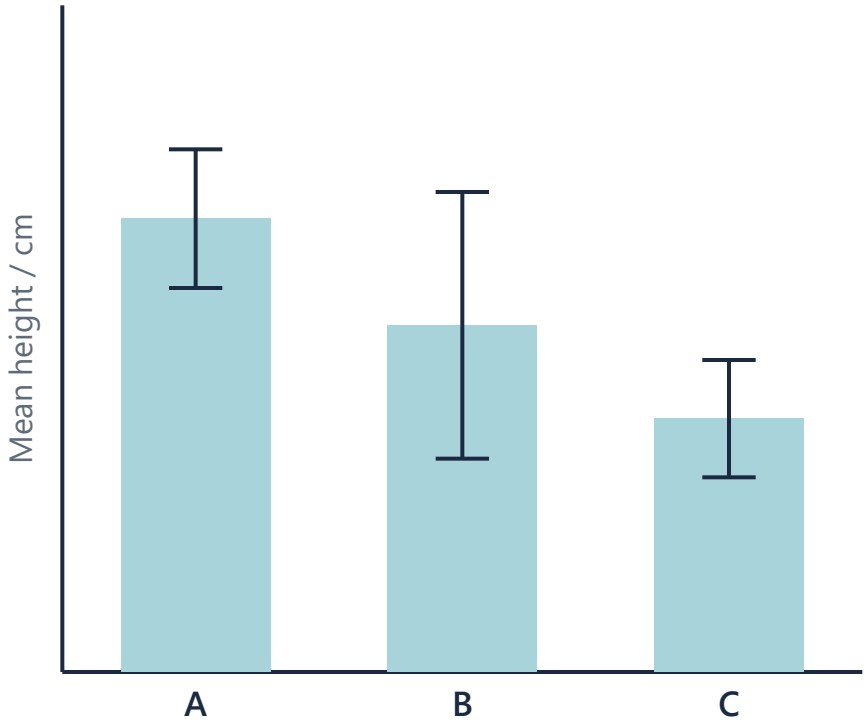
Outliers

The value 12 pulls the mean up more than the median. Check an outlier for a mistake before leaving it out, and say if you do.



Stefan (Diskussion). Original uploader, Public domain

Standard deviation and error bars



SD measures spread around the mean. About 68% of values lie within 1 SD. Error bars show +/- 1 SD.

Reading the bars

A and B overlap a lot: the difference may not be significant. A and C do not overlap: a real difference is more likely. Test to be sure.



StockyPics, CC0

TRY IT

Interpret error bars

Plants with fertiliser: mean 24 cm, SD 2 cm. Plants without: mean 18 cm, SD 6 cm.

1. Which group is more variable?
2. Do the error bars (+/- 1 SD) overlap?
3. Can you be sure fertiliser increased height?

> Hint: Find mean +/- SD for each group: 22 to 26, and 12 to 24.

The answer is on the next slide.

Gerda Arendt, CC0

ANSWER

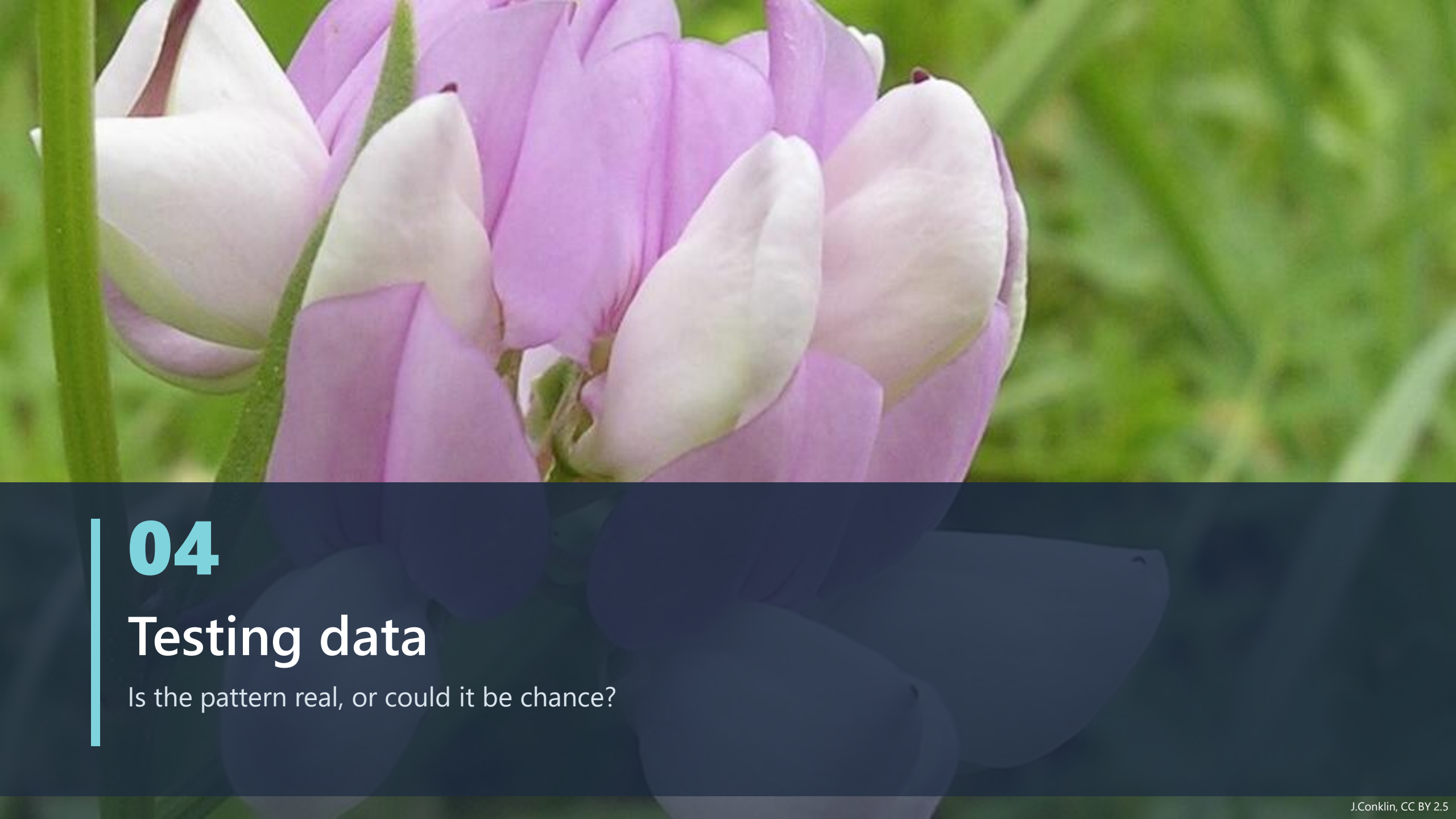
Interpret error bars

1. Without fertiliser: larger SD, so heights vary more
2. Yes: 22 to 26 cm overlaps 12 to 24 cm
3. No: the difference may be due to chance. A t-test would show whether it is significant

Compare your working with each step.

Key point

Overlapping bars suggest, but do not prove, that a difference is not significant.



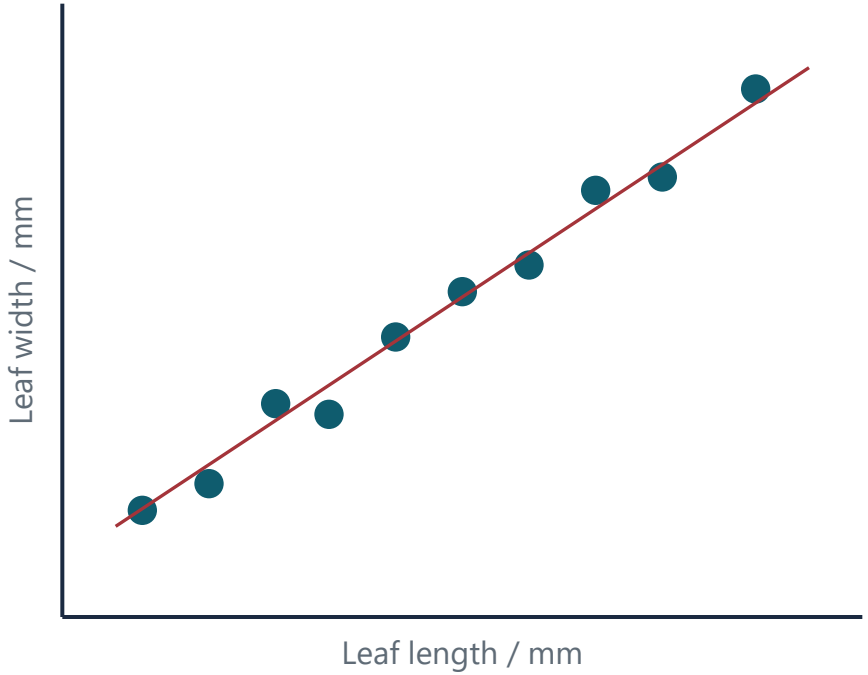
04

Testing data

Is the pattern real, or could it be chance?

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Correlation



r runs from -1 (perfect negative) through 0 (none) to +1 (perfect positive). Here r = 0.98.

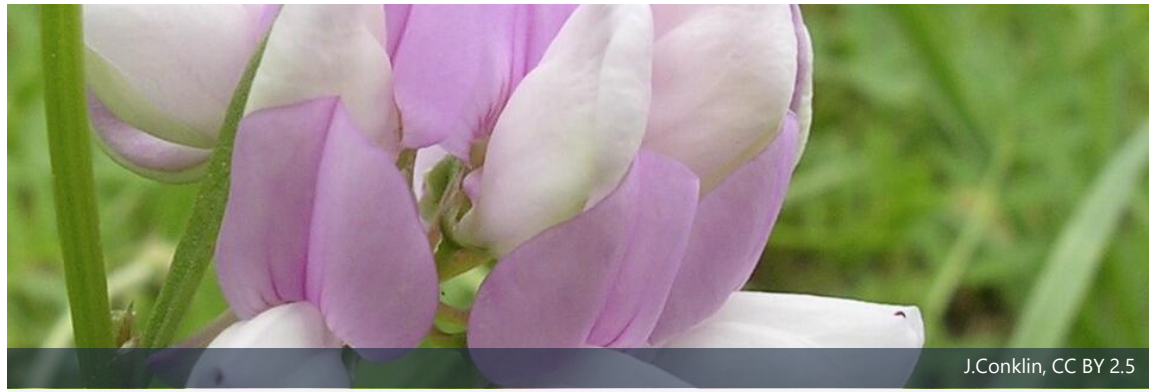
$r = 0.8$, so $r^2 = 0.64$

64% of the variation in one variable is linked to the other. Correlation still does not show cause.



Aethonatic, CC0

Choosing a statistical test



J.Conklin, CC BY 2.5

Chi-squared

Counts in categories. Observed against expected: do offspring fit 3:1? Are two species found together (C4.1)?



Florida Memory, Public domain

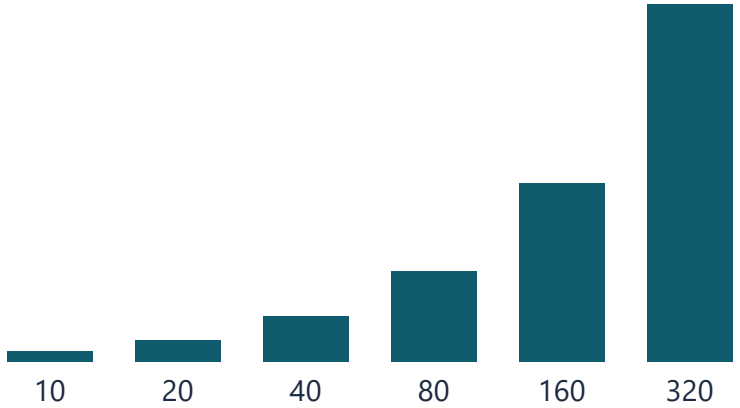
t-test

Compares two means of measured data: resting pulse of athletes and non-athletes. Common in IAs.

1. State the null hypothesis
2. Compare with the critical value at p = 0.05
3. Larger? Reject the null hypothesis

Logarithms and log scales

- On a log10 scale each step is **ten times** bigger: 1, 10, 100, 1000
- Exponential growth becomes a **straight line** (C4.1.8)
- pH is a log scale: pH 5 has ten times the hydrogen ions of pH 6



Duckweed fronds, every 3 days

Activity

Plot these on normal and on log paper. Which gives a straight line, and why?



Kostka Martin, CC0

TRY IT

Which test?

Chi-squared or t-test?

1. Mean pulse rate of 20 athletes and 20 non-athletes.
2. Purple and white flowers in offspring, against a 3:1 prediction.
3. Presence of two snail species in 50 quadrats.
4. Mean root length of seedlings in two soils.

> Hint: Counts in categories: chi-squared. Two means: t-test.

The answer is on the next slide.

J.Conklin, CC BY 2.5

ANSWER

Which test?

1. t-test: two means
2. Chi-squared: observed against expected counts
3. Chi-squared: association from presence and absence counts
4. t-test: two means

Compare your working with each step.

Key point

Ask first: counts in categories, or measurements with a mean?

Check that you can

1. Convert m, mm, µm and nm, and use standard form
2. Give sensible significant figures
3. Calculate percentage change
4. Calculate magnification and actual size
5. Find a rate from a gradient or tangent
6. Find mean, median, mode and range
7. Interpret SD, error bars and r
8. Choose chi-squared or a t-test, and use log scales



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Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. A mass changes from 5.0 g to 4.6 g. Calculate the percentage change.

2. An organelle 2 µm long appears 10 mm long. Calculate the magnification.

3. Which statistical test compares two means?

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Russet potato cultivar with sprouts.jpg – ZooFari – Public domain

Veac102fS2Atop-TEM of Chlamydomonas reinhardtii CC-2937 cells.png – Mohammad Moniruzzaman, Maria P. Erazo-Garcia, Frank O. Aylwa – CC BY 4.0

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